

OWENSBORO-DAVISS COUNTY AP, KY, USA

WMO: 724237

Lat: 37.750N

Lon: 87.167W

Elev: 403

StdP: 14.48

Time Zone: -6.00 (NAC)

Period: 06-19

WBAN: 53803

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 9.1	(c) 14.2	(d) -1.0	(e) 5.3	(f) 12.4	(g) 3.5	(h) 6.7	(i) 15.4	(j) 26.3	(k) 42.9	(l) 23.8	(m) 41.4	(n) 6.4	(o) 310	(p) 0.465

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	94.8	77.5	92.6	76.9	90.7	76.2	80.8	90.9	79.6	89.4	78.2	86.9	8.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.1	148.2	86.7	76.9	142.2	85.3	75.7	136.7	83.8	44.6	90.8	43.2	89.3	41.8	86.6	86.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 22.7	(b) 19.2	(c) 17.1	(e) 3.4	(f) 98.2	(g) 5.5	(h) 3.6	(i) -0.5	(j) 100.9	(k) -3.7	(l) 103.0	(m) -6.8	(n) 105.0	(o) -10.8	(p) 107.6		
(4) 22.7	(5) 19.2	(6) 17.1	(7) 3.4	(8) 98.2	(9) 5.5	(10) 3.6	(11) -0.5	(12) 100.9	(13) -3.7	(14) 103.0	(15) -6.8	(16) 105.0	(17) -10.8	(18) 107.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.2	33.4	37.2	48.2	58.9	68.8	77.1	78.9	78.0	71.5	59.1	46.7	39.4
	DBStd	18.37	11.49	12.09	11.00	9.33	7.86	5.04	4.77	4.78	6.94	9.40	9.52	10.04
	HDD50	1658	527	385	170	29	0	0	0	0	0	22	171	354
	HDD65	4175	980	778	529	224	59	2	0	0	20	235	553	795
	CDD50	4654	11	27	114	296	582	814	895	869	646	304	71	25
	CDD65	1695	0	1	8	40	175	366	431	404	216	52	2	0
	CDH74	16458	0	2	64	354	1492	3619	4397	4007	2002	508	13	0
	CDH80	6760	0	0	7	66	500	1552	1948	1722	812	152	1	0
Wind	WSAvg	7.5	9.0	8.9	9.1	9.0	7.8	6.9	5.6	5.3	5.8	7.1	7.6	8.5
Precipitation	PrecAvg	46.9	3.2	3.7	4.8	4.6	4.6	3.7	3.8	3.4	3.5	2.9	4.0	3.8
	PrecMax	58.5	7.7	11.0	17.7	12.2	11.9	8.4	8.4	11.0	7.4	7.6	9.9	7.3
	PrecMin	27.7	0.6	0.6	0.8	1.1	1.4	1.0	1.1	0.4	0.4	0.1	1.0	0.9
	PrecStd	6.8	1.8	2.2	2.9	2.5	2.4	1.8	1.8	2.4	2.0	1.7	2.0	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.9	71.2	81.0	85.3	91.2	95.7	99.4	98.9	95.3	89.4	75.8	69.0
		MCWB	60.5	61.9	66.2	67.9	74.7	78.2	81.8	79.2	74.0	72.5	62.3	62.7
	2%	DB	60.2	66.0	74.9	82.1	87.7	93.2	94.3	94.3	92.0	84.3	70.6	63.5
		MCWB	56.6	59.3	62.8	66.4	71.8	76.3	79.1	77.3	74.2	68.5	60.7	59.6
	5%	DB	55.8	61.5	70.3	78.6	85.5	91.1	92.0	91.4	89.0	80.0	66.1	59.3
		MCWB	52.6	55.9	60.7	64.9	71.2	75.8	78.5	76.9	73.5	66.1	58.1	55.8
	10%	DB	50.8	56.4	65.3	74.5	82.9	88.7	89.5	88.8	84.9	75.2	61.9	55.1
		MCWB	46.7	51.1	56.9	62.5	69.9	74.5	77.2	75.8	71.3	64.6	55.2	51.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.6	63.8	67.4	71.5	76.8	80.6	83.0	82.3	79.6	74.7	66.4	64.0
		MCDB	64.2	69.0	77.6	81.3	87.1	91.1	93.9	93.0	84.8	85.3	71.9	67.3
	2%	WB	57.4	60.5	64.5	69.2	74.8	79.2	81.2	80.3	76.8	72.3	63.0	60.4
		MCDB	60.0	64.9	72.6	78.0	84.3	89.9	91.5	90.6	86.7	81.0	67.8	62.8
	5%	WB	53.2	57.0	62.0	66.9	73.2	77.8	80.0	78.9	75.3	69.2	59.5	56.1
		MCDB	55.2	60.6	68.4	74.3	82.3	87.3	89.7	88.2	85.8	74.8	64.4	58.2
	10%	WB	47.1	50.9	58.6	64.6	71.4	76.4	78.5	77.5	73.5	66.2	55.9	51.4
		MCDB	49.9	55.5	64.5	71.7	79.8	85.7	87.0	85.4	81.6	72.9	61.1	54.5
Mean Daily Temperature Range	5% DB	MDBR	15.8	16.8	18.3	20.6	18.7	18.8	18.2	19.0	20.5	21.1	19.0	14.9
		MCDBR	21.5	22.6	23.6	24.6	21.7	21.6	21.1	21.9	24.5	25.6	24.2	20.5
		MCWBR	18.3	17.7	14.7	12.6	9.4	8.6	8.7	9.1	9.8	12.0	15.3	17.1
	5% WB	MCDBR	19.7	20.9	20.9	19.9	19.1	19.5	19.5	19.6	20.1	21.2	19.6	18.6
		MCWBR	18.6	17.8	14.3	12.4	9.1	8.7	9.0	8.9	9.0	11.5	15.4	17.1
Clear-Sky Solar Irradiance	taub	0.317	0.326	0.365	0.407	0.434	0.451	0.476	0.486	0.414	0.363	0.331	0.310	
	taud	2.460	2.411	2.365	2.252	2.225	2.197	2.141	2.103	2.298	2.431	2.472	2.486	
	Ebn at Noon	275	287	284	277	270	265	257	251	265	270	267	270	
	Edh at Noon	26	31	36	43	45	46	48	49	38	30	26	24	
All-Sky Solar Radiation	RadAvg	632	849	1189	1576	1765	2008	1921	1774	1472	1101	755	544	
	RadStd	67	105	110	95	156	121	121	96	112	117	81	52	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (31 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+207	+146

Nomenclature: See separate page